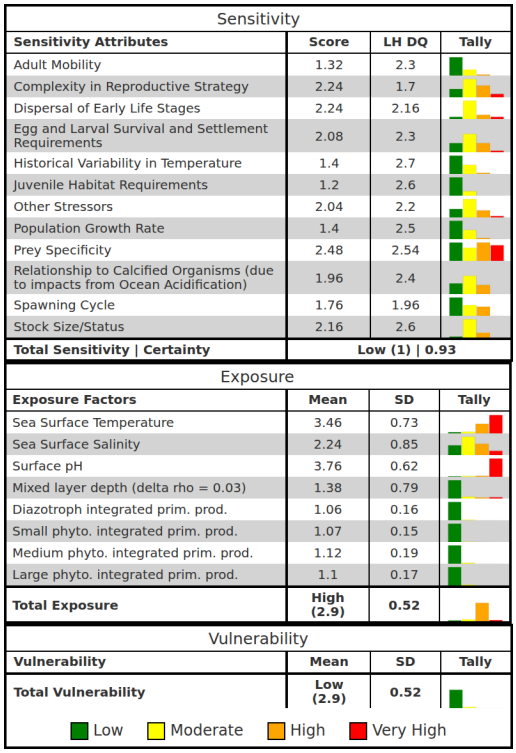
Atlantic Menhaden (*Brevoortia tyrannus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.32	2.3	
Complexity in Reproductive Strategy	2.24	1.7	
Dispersal of Early Life Stages	2.24	2.16	
Egg and Larval Survival and Settlement Requirements	2.08	2.3	
Historical Variability in Temperature	1.4	2.7	
Juvenile Habitat Requirements	1.2	2.6	
Other Stressors	2.04	2.2	
Population Growth Rate	1.4	2.5	
Prey Specificity	2.48	2.54	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.96	2.4	
Spawning Cycle	1.76	1.96	
Stock Size/Status	2.16	2.6	
Total Sensitivity | Certainty	**Low (1) | 0.93**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	3.46	0.73	
Sea Surface Salinity	2.24	0.85	
Surface pH	3.76	0.62	
Mixed layer depth (delta rho = 0.03)	1.38	0.79	
Diazotroph integrated prim. prod.	1.06	0.16	
Small phyto. integrated prim. prod.	1.07	0.15	
Medium phyto. integrated prim. prod.	1.12	0.19	
Large phyto. integrated prim. prod.	1.1	0.17	
Total Exposure	**High (2.9)**	**0.52**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.9)**	**0.52**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic Menhaden. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	Very High	Moderate	Decrease
Vulnerability	Moderate	Low	Decrease
Directional Effect	Positive	Positive	None

Stocks

Atlantic Menhaden is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

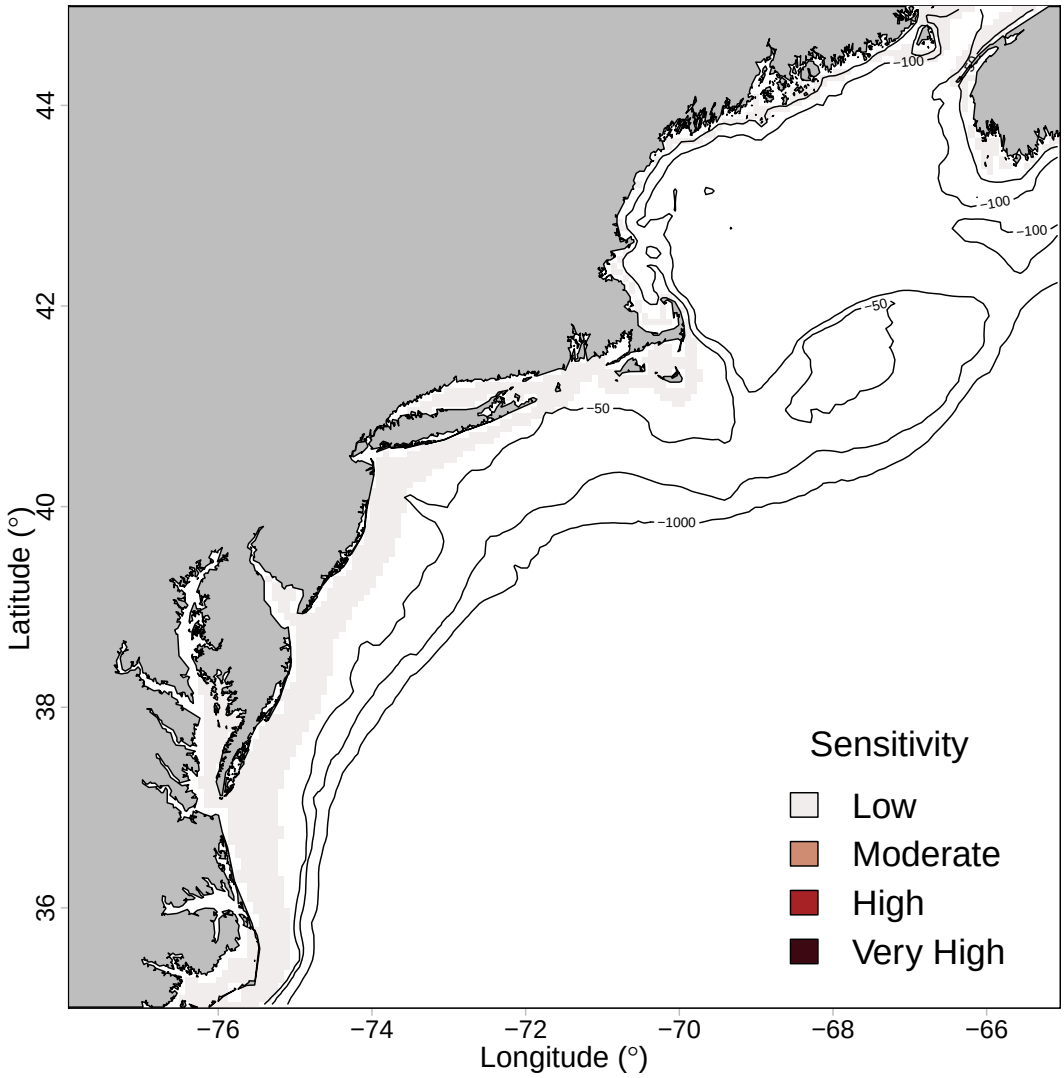


Figure 2: Atlantic Menhaden Sensitivity on the Northeast Shelf

Atlantic menhaden received a **Low** sensitivity score, with a certainty of 93% from bootstrap analysis (Figure 1). The greatest attribute score was Prey Specificity (2.5) (Figure 1). Juveniles and adults both consume zooplankton, but are generalist feeders that are not likely to be impacted by mismatches (Rogers and Van Den Avyle 1989).

The previous Climate Vulnerability Assessment also gave menhaden a Low sensitivity score (Hare et al. 2016). The highest attribute in that assessment was Early Life History Requirements because late larval and juvenile life stages depend on estuaries (Able 2005). None of the estuarine habitats that these life stages rely on were classified as vulnerable by the habitat climate vulnerability assessment (Guyant et al. 2022).

Exposure:

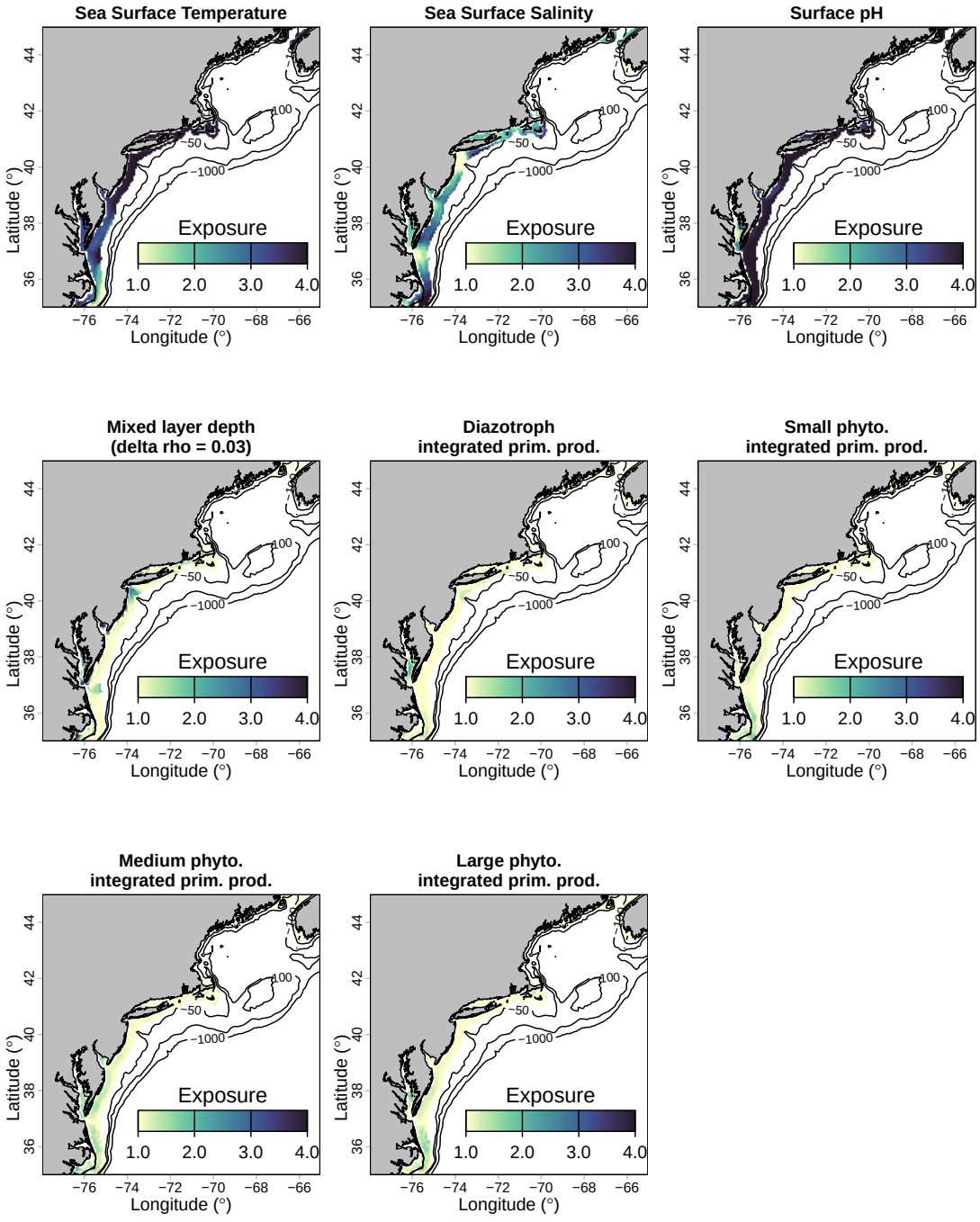


Figure 3: Maps of each Exposure Factor for Atlantic Menhaden. Exposure was calculated between 2014-2023 and 2025-2035

The mean Exposure score for Atlantic menhaden was 2.9 (+/- 0.52), giving the species a **High** mean score (Figure 1). Exposure was High throughout most of the species' range, however Low scores within

Chesapeake Bay and off the coast of Virginia/North Carolina drove down the average score (Figure 4). High scores were driven by high exposure to changes in sea surface temperature and pH (Figure 3).

In the previous Assessment, menhaden received a Very High exposure score due to exposure to changes in sea surface temperature, ocean acidification, and air temperatures (Hare et al. 2016).

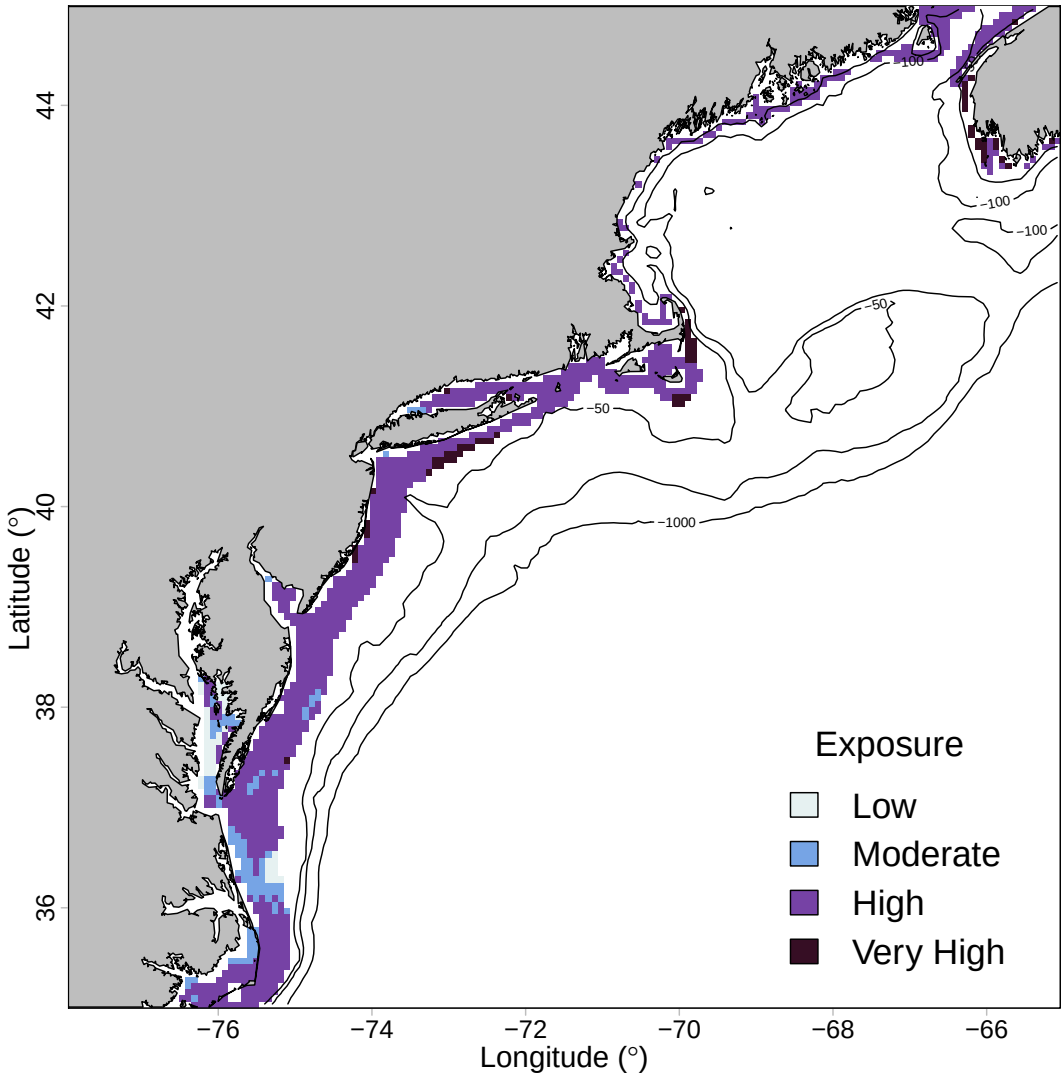


Figure 4: Total Exposure on the Northeast Shelf for Atlantic Menhaden. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

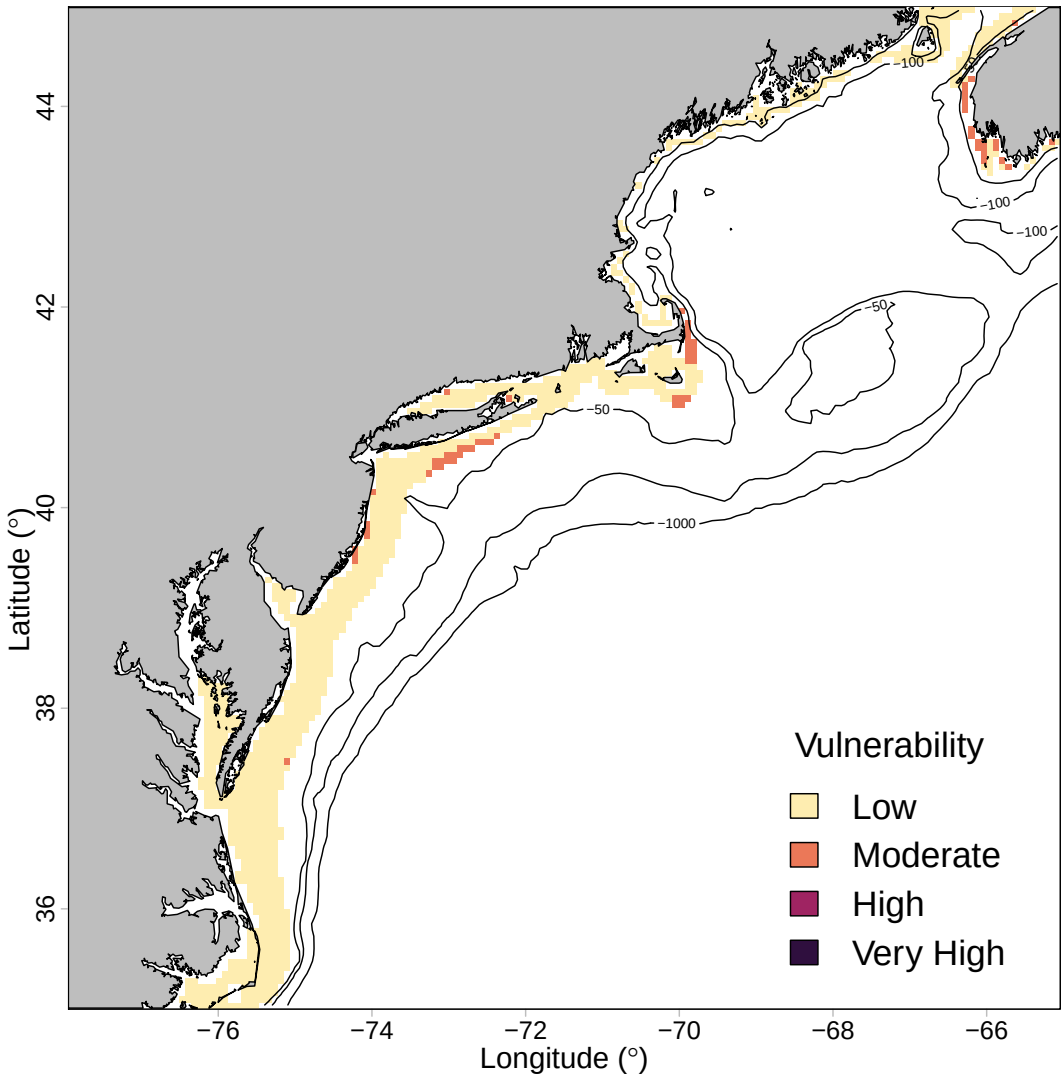


Figure 5: Vulnerability on the Northeast Shelf for Atlantic Menhaden. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean total Vulnerability was **Low** for Atlantic menhaden (2.9 +/- 0.52) (Figure 1). Scores were low throughout Menhaden's range, with some areas of moderate vulnerability off the coast of Long Island and south of Cape Cod (Figure 5).

In the previous assessment, menhaden received a Moderate vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality (Figure 1). Complexity in Reproductive Strategy and Spawning Cycle were identified as data gaps (mean attribute

data quality score < 2) (Figure 1).

Model Confidence Score:

The menhaden ensemble model received a model confidence score of 2 (+/- 0.4), indicating moderate confidence in the model. Experts stated that the model's general distribution trends matched expectations, but missed recent northward expansions. The ensemble model's AUC was 0.84, indicating good performance.

Additional Metrics

Distribution Shifts:

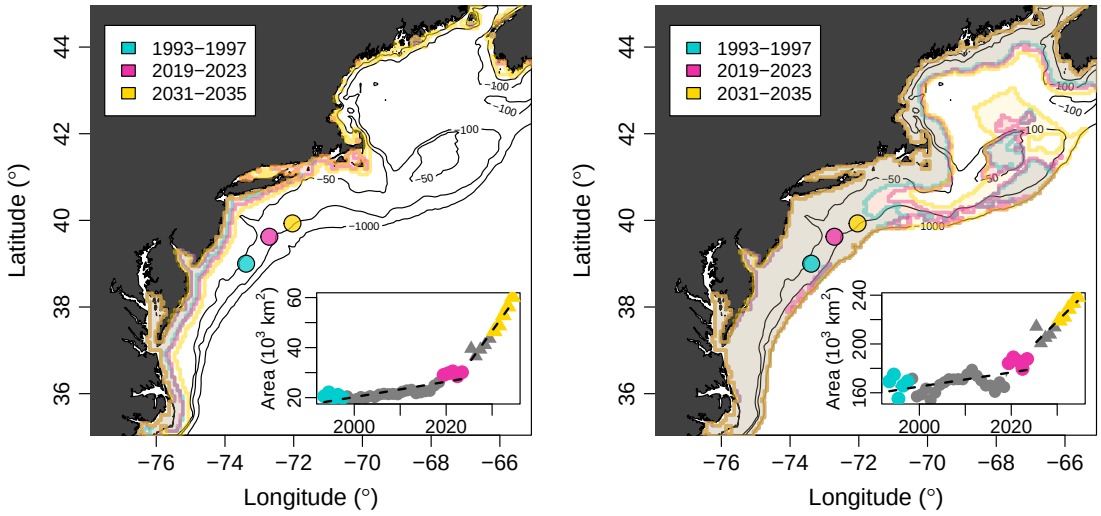


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic Menhaden. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Menhaden habitat area expanded north and east significantly between 1993 and 2023 and the weighted centroid shifted 90 km northeast. Menhaden distributions shifted further offshore and further north into the Gulf of Maine (Figure 6).

In the next ten years, these trends are predicted to continue. Habitat area is predicted to significantly increase as menhaden distributions continue to expand offshore and northward into the Gulf of Maine (Figure 6). The weighted centroid is predicted to shift a further 67 km east-northeast.

In the previous Assessment, Atlantic menhaden received a High Distributional Vulnerability Rank due to wide larval dispersal and large seasonal migrations (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	20,795	168,520	8.85	9.46
2019-2023	90.5 km NE	29,647 (+8,852)	186,690 (+18,170)	8.85 (0)	9.78 (+0.32)
2031-2035	67 km ENE	53,280 (+23,633)	230,160 (+43,469)	8.72 (-0.12)	9.70 (-0.08)

Directional Effect:

Atlantic menhaden are estimated to be **positively** impacted by climate change. Expert certainty was 58% from bootstrap analysis. Menhaden distributions have been expanding northward (Morano 2025) likely due to warming and increased spawning activity. These results are similar to those from the 2016 Climate Vulnerability Assessment, albeit with decreased certainty (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Atlantic menhaden are estuary dependent species, and their habitats may be impacted by future development or trawling and dredging activities (Rogers and Van Den Avyle 1989). Atlantic menhaden distribution is expanding northward, likely to due warmer water temperatures, although other local ecological factors may also influence distribution (Morano 2025). Recruitment of Atlantic menhaden larvae to nursery habitats appears to be driven primarily by larval supply and the oceanographic properties driving transport, as well as nursery habitat quality, rather than specific settlement cues (Quinlan, Blanton, and Werner 1999). Climate change could impact these transport mechanisms or the quality of nursery habitats. In a study modeling menhaden recruitment trends in Chesapeake Bay and Southern New England, a broad-scale climate driver, the Atlantic Multidecadal Oscillation (AMO) was one of the best predictors of recruitment (Buchheister et al. 2016). Midway et al. (2020) used hierarchical models to determine significant effects of age 1 Atlantic menhaden growth caused by the AMO (negative effect), easterly (positive effect) and northerly (negative effect) wind velocity. Turner (2017) investigated declines in size-at-age for age 3-5 Atlantic menhaden and attributed these size changes to coastal and ocean warming. This study also predicted future reductions in size-at-age as ocean temperatures rise.

Life History Synopsis

Atlantic menhaden inhabit coastal and estuarine waters from Nova Scotia to Florida (Rogers and Van Den Avyle 1989). They are a key forage species in the North and South Atlantic, and are consumed by a wide range of predators, including striped bass, bluefish, and weakfish in the North Atlantic (Atlantic States Marine Fisheries Commission 2025). Menhanden spawning peaks in December - February (SouthEast Data, Assessment, and Review 2025). Spawning was originally thought to be concentrated off the coast of the Carolinas following the fall migration. However, more recent analysis of larval distributions suggests that spawning occurs throughout the range, as far north as Long Island (Nelson, Ingham, and Schaaf 1977) (Simpson et al. 2017). Menhaden spawning periods coincide with presence of phytoplankton and zooplankton, the dominant food sources of larval menhaden (Rogers and Van Den Avyle 1989). Eggs

are buoyant and reliant on transportation by wind and tidal currents towards nursery estuary habitats, where they hatch in a few days (SouthEast Data, Assessment, and Review 2025). After around 30 days in the larval stage, juveniles move to upper portions of estuaries including river shoals and the heads of small tidal creeks, where they congregate over vegetation in open water (Rogers and Van Den Avyle 1989). Young juvenile menhaden consume individual zooplankton; this reliance on zooplankton decreases as menhaden move from open waters to marshes, although adult menhaden also consume zooplankton (Rogers and Van Den Avyle 1989). Adults prefer temperatures near 18°C and generally remain within the top 100m of the water column (SouthEast Data, Assessment, and Review 2025). After the winter spawning season, adults migrate inshore and north following spawning grounds in the Carolinas to the Gulf of Maine (Rogers and Van Den Avyle 1989). Atlantic menhaden reach maturity around 2 years of age and a maximum age is listed at 10 years, although individuals older than 6 years of age are rarely caught in the fishery (SouthEast Data, Assessment, and Review 2025). Menhaden in the Northwest Atlantic are managed as a single stock unit by the Atlantic States Marine Fisheries Commission. The most recent stock assessment update in 2025 used ecological reference points to determine stock status. As of 2025, the stock is not overfished and overfishing is not occurring (Atlantic States Marine Fisheries Commission 2025).

Supplemental Scores

Exposure Timeseries

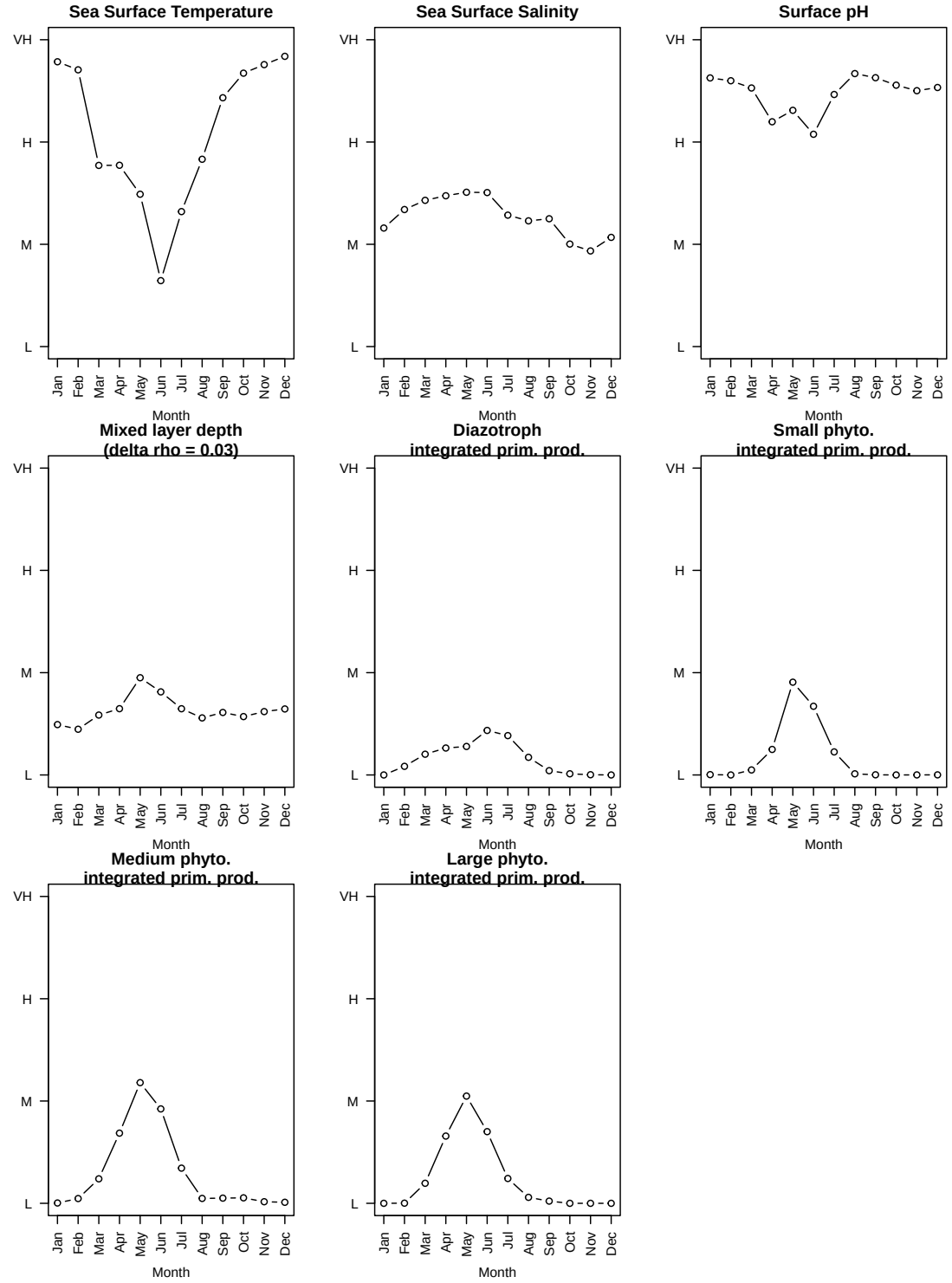


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic Menhaden. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure scores ranged from Low to High. Scores were Low in July and High from September through February (Figure 8). These patterns were driven by changes in sea surface temperature and pH (Figure 7).

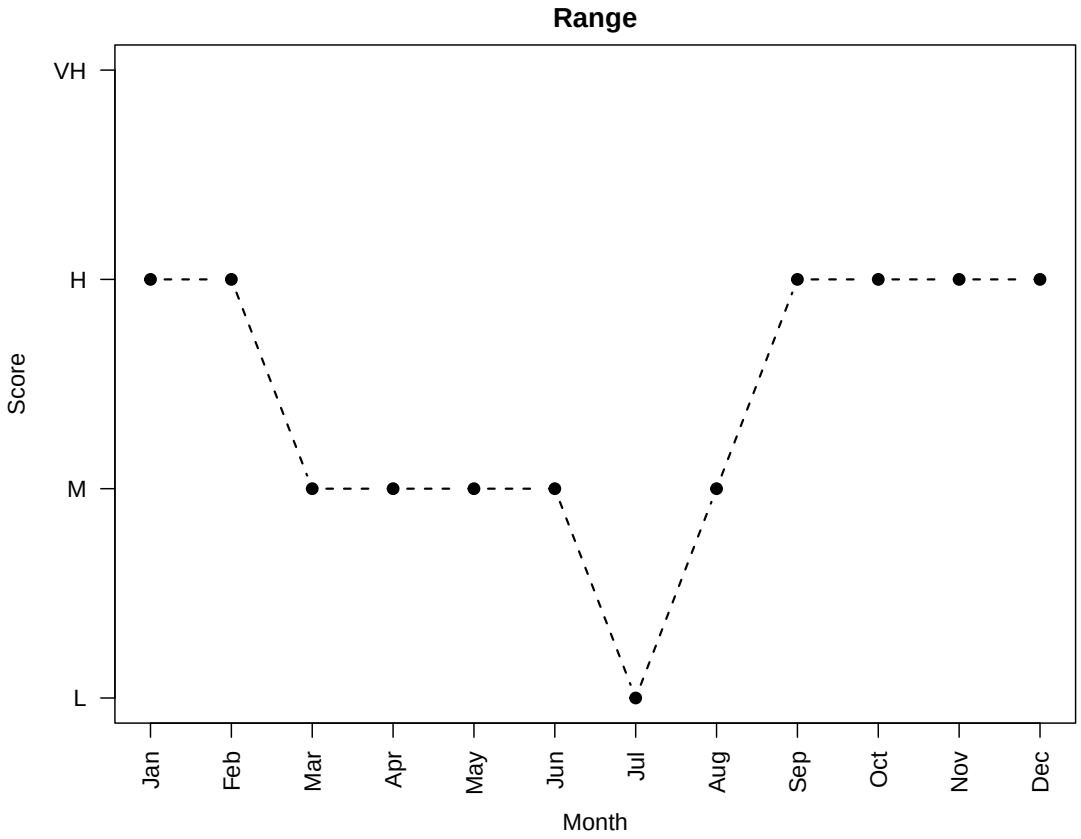


Figure 8: Timeseries of Total Exposure for Atlantic Menhaden

Exposure due to Significant Factors

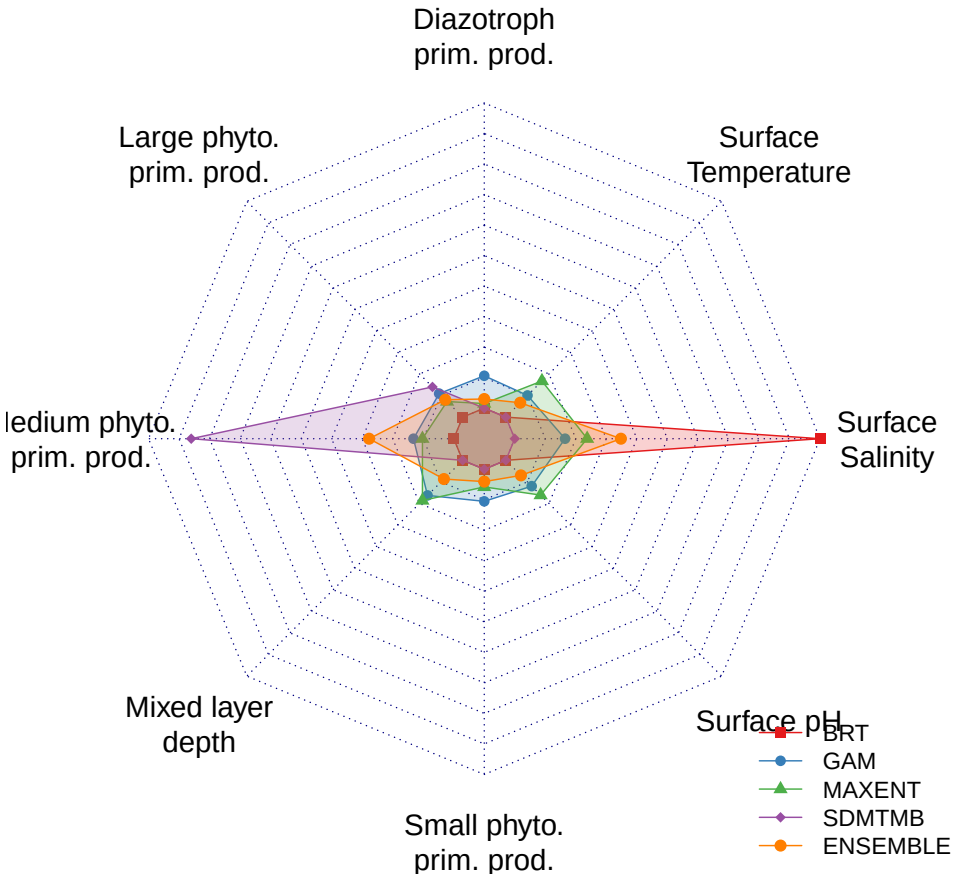


Figure 9: Dynamic Factors driving Atlantic Menhaden distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface salinity and medium phytoplankton primary productivity were important drivers of species distributions (Figure 9). Menhaden are estuary-dependent species, and consume phytoplankton and zooplankton (Rogers and Van Den Avyle 1989). Only changes in sea surface salinities resulted in high exposures. Therefore, exposure scores when only important factors were considered was Low (1 +/- 0) (Figure 10; Figure 11).

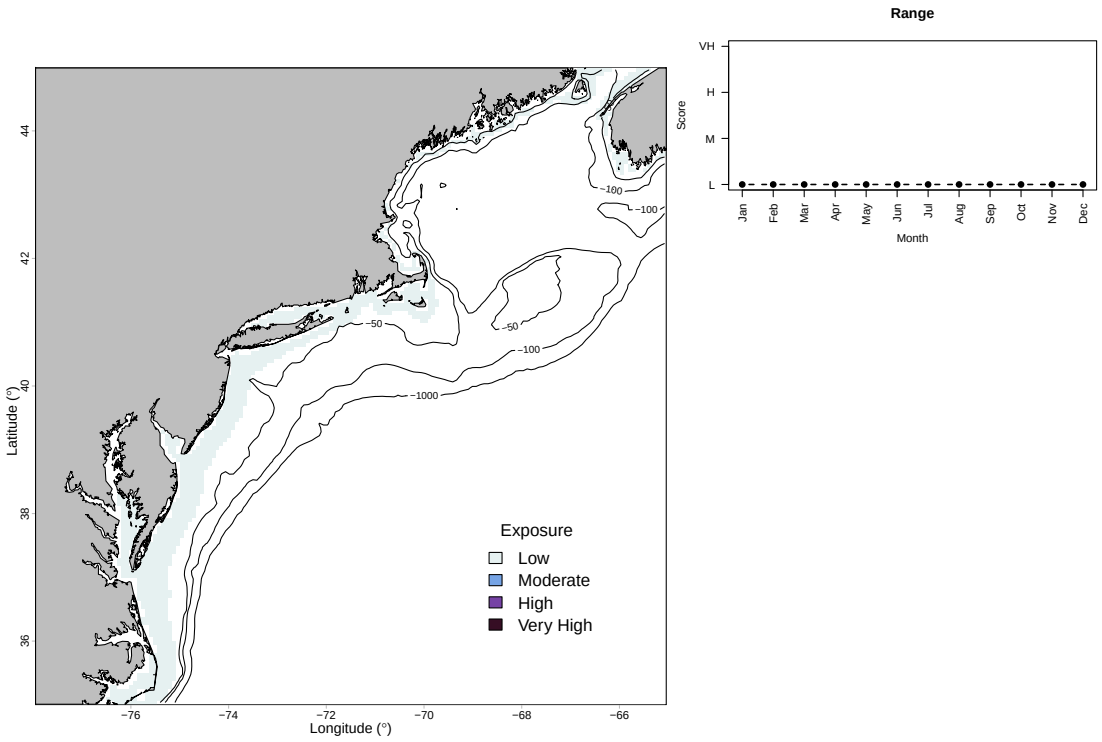
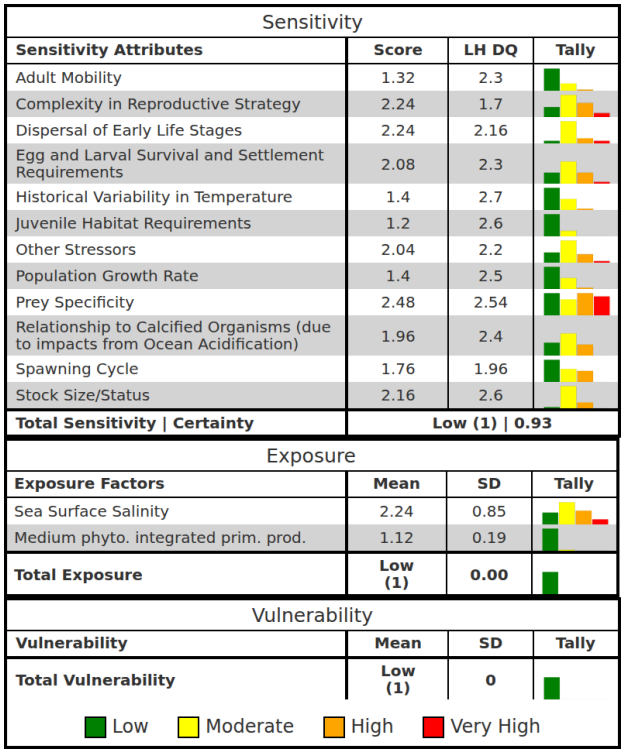


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic Menhaden distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.32	2.3	
Complexity in Reproductive Strategy	2.24	1.7	
Dispersal of Early Life Stages	2.24	2.16	
Egg and Larval Survival and Settlement Requirements	2.08	2.3	
Historical Variability in Temperature	1.4	2.7	
Juvenile Habitat Requirements	1.2	2.6	
Other Stressors	2.04	2.2	
Population Growth Rate	1.4	2.5	
Prey Specificity	2.48	2.54	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.96	2.4	
Spawning Cycle	1.76	1.96	
Stock Size/Status	2.16	2.6	
Total Sensitivity | Certainty	**Low (1) | 0.93**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Salinity	2.24	0.85	
Medium phyto. integrated prim. prod.	1.12	0.19	
Total Exposure	**Low (1)**	**0.00**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (1)**	**0**	

Figure 11: As in Figure 1 for Atlantic Menhaden, but with only significant Exposure factors and the resulting Vulnerability Scores.

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